

Vitruvian Recurrent Unit

Document 2: Mathematical Formalization

Cell Architecture · Geometric Derivation · Initial Gradient Results

Dylan Michael Scott · Horizon Tech · March 2026

$\phi = 4/\pi = 1.273239544735163$

1. Purpose of This Document

Document 1 (Origin) established the geometric derivation of $\phi = 4/\pi$ from Da Vinci's Vitruvian Man and confirmed it as a natural proportional attractor through field simulation. Document 2 formalizes that finding into a precise mathematical cell definition, states the core theoretical claims as falsifiable conjectures, and records the initial empirical validation through v2 gradient experiments at $\text{seq_len}=1000$. This document is the bridge between geometric intuition and experimental science.

2. The DPPU Cell — Formal Definition

The Dynamic Pi-Phi Processing Unit (DPPU) is a recurrent cell defined by a single modification to the standard Elman RNN: the recurrent weight path is scaled by the geometric constant $\phi = 4/\pi$.

Standard Elman RNN:

$$h_t = \tanh(w_x * x_t + w_h * h_{t-1} + b)$$

DPPU Cell:

$$h_t = \tanh(w_x * x_t + \phi * w_h * h_{t-1} + b)$$

Where $\phi = 4/\pi = 1.273239544735163$ is a fixed scalar constant derived from the dual-center geometry of the Vitruvian Man. It is not a learned parameter, not a gate, and not a normalization layer. It adds zero parameters to the architecture.

2.1 Parameter Count

Component	DPPU-RNN	Vanilla-RNN	LSTM
Input projection W_x	input × hidden	input × hidden	4 × (input × hidden)
Recurrent weight W_h	hidden × hidden	hidden × hidden	4 × (hidden × hidden)
Bias b	hidden	hidden	4 × hidden
Phi constant	fixed (0 params)	—	—
Gates	none	none	3 sigmoid gates
Total (hidden=32)	1,121	1,121	4,513

Total (hidden=64)	4,289	4,289	17,217
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Table 1. Parameter comparison at two hidden dimensions. DPPU is identical to Vanilla — phi costs nothing.

3. Derivation of $\phi = 4/\pi$

The Vitruvian Man encodes a dual-center geometry: one center at the navel (the circle's center) and one at the base of the pelvis (the square's center). The ratio between the circumscribed circle radius and the inscribed square half-side in this canonical figure produces a natural proportional constant.

Geometric derivation:

Circle radius $r = 1$ (normalized)

Square side $s = \pi/2$ (from Vitruvian proportions)

Square half-side $= \pi/4$

$\phi = r / (\pi/4) = 4/\pi = 1.273239\dots$

This is the ratio at which circular and rectilinear geometry are in proportion — the boundary between oscillatory and linear dynamics. In the field simulation recorded in Document 1, this value emerged as the natural attractor of the $\pi_{\text{dynamic}} \times \phi_{\text{dynamic}}$ product function at the critical transition point Δ^* , confirming it is not an arbitrary choice but a geometric fixed point.

3.1 Dynamic Pi and Phi Functions

The field simulation from which ϕ was confirmed operates on two dynamic functions that interpolate between classical and limiting values as a function of the field tension parameter Δ :

$\pi_{\text{dynamic}}(\Delta) = 4 - (4 - \pi_{\text{classical}}) * \exp(-\Delta)$

$\phi_{\text{dynamic}}(\Delta) = \phi_{\text{classical}} * \exp(-\Delta) + (1 - \exp(-\Delta))$

$\Omega(\Delta) = (\pi_{\text{dynamic}} * \phi_{\text{dynamic}}) / (1 + \Delta)$

At $\Delta = 0$: $\pi_{\text{dynamic}} \rightarrow 4$, $\phi_{\text{dynamic}} \rightarrow \phi_{\text{classical}} = 4/\pi$. At $\Delta \rightarrow \infty$: $\pi_{\text{dynamic}} \rightarrow \pi_{\text{classical}}$, $\phi_{\text{dynamic}} \rightarrow 1$. The critical transition $\Delta^* \approx 0.816$ is where Ω is maximized — the operating point of the DPPU cell.

4. Core Theoretical Claims

The DPPU framework rests on four falsifiable conjectures. Documents 3 through 9 progressively test each one.

Conjecture 1 — Gradient Stability:

Scaling the recurrent weight path by $\phi = 4/\pi$ biases the effective spectral radius of W_h toward a stable geometric fixed point, preventing both vanishing and exploding gradients across arbitrarily long sequences.

Conjecture 2 — Parameter Efficiency:

A DPPU cell with n parameters achieves performance comparable to an LSTM with $4n$ parameters on sequence prediction tasks, because ϕ provides geometric gradient stability equivalent to what LSTM achieves through learned gating.

Conjecture 3 — Spectral Attractor:

During training, gradient descent drives the spectral radius of W_h toward $\phi/2 = 2/\pi \approx 0.6366$, so the effective recurrent spectral radius $\phi \times \rho(W_h)$ converges toward $\phi^2/2 = 8/\pi^2 \approx 0.8106$.

Conjecture 4 — Scaling Invariance:

The $\phi^2/2$ effective spectral radius is stable independent of sequence length. DPPU performance should improve or hold as sequences grow longer, while vanilla RNN performance degrades.

5. Initial Empirical Validation — v2 Gradient Results

The first controlled experiment (v2) trained DPPU and Vanilla-RNN side by side on sine wave next-step prediction at $\text{seq_len}=200$, establishing a clean baseline. The v3 experiment extended to $\text{seq_len}=1000$, producing the first evidence of gradient advantage.

Experiment	Seq Len	DPPU Loss	Vanilla Loss	DPPU Grad	Vanilla Grad	Grad Ratio
v2 baseline	200	~stable	~stable	—	—	both stable
v3 gradient	1000	comparable	comparable	stronger	baseline	1.57×

Table 2. v2 and v3 results. At $\text{seq_len}=1000$, DPPU gradient signal is 1.57× stronger than Vanilla at identical parameter count — first empirical confirmation of Conjecture 1.

The 1.57× gradient strength ratio at $\text{seq_len}=1000$ is the seed result of the entire experimental series. It established that ϕ is doing real gradient work — not a cosmetic architectural change — and justified all subsequent experiments.

6. Reference Cell Implementation

The canonical DPPU cell as first implemented. All subsequent versions (v3 through v23) modify this foundation — adding carry fields, registers, or training infrastructure — but the core ϕ scaling on the recurrent path is preserved throughout.

```
class DPPUCell(nn.Module):
    def __init__(self, input_dim, hidden_dim):
        super().__init__()
        self.phi = 4.0 / math.pi # 1.273239544735163
        self.W_x = nn.Linear(input_dim, hidden_dim, bias=True)
        self.W_h = nn.Linear(hidden_dim, hidden_dim, bias=False)
    def forward(self, x, h):
        # phi scales recurrent weight path — no gates, no extra params
        return torch.tanh(self.W_x(x) + self.phi * self.W_h(h))
```

7. Why This Is Not Just a Scaled RNN

A naive reading of the cell definition might suggest that ϕ is simply a learning rate adjustment on the recurrent path — that it could be absorbed into the weight initialization or optimizer settings. This reading is incorrect for three reasons.

First:

ϕ is a fixed geometric constant, not a learned scalar. It does not change during training. This means it cannot be absorbed into W_h through learning — it permanently biases the geometry of the recurrent weight space.

Second:

The spectral attractor conjecture predicts that ϕ drives W_h to a specific spectral radius ($2/\pi$) through the gradient descent process itself. This is a prediction about what W_h converges to — not just how fast it trains. A learned scalar cannot produce this effect because it would simply be incorporated into W_h .

Third:

The field simulation confirms ϕ as the natural attractor of the critical transition in the dynamic π - ϕ system. The constant is derived from geometry, not tuned from data. This geometric grounding is what separates the DPPU framework from empirical hyperparameter search.

8. Position in Experimental Series

Document	Label	Content
1	Origin (DPPU)	Da Vinci geometry — derivation of $\phi = 4/\pi$, field simulation
2	Formalization (this)	Cell definition, conjectures, v2-v3 gradient results
3	Progress v2 (DPPU)	v2–v5 experiments, gradient strength confirmation
4	LSTM Comparison (DPPU)	v6 three-way comparison — DPPU beats LSTM at 3.7x fewer params
5	Stress Test (DPPU)	seq_len=5000 full BPTT — DPPU achieves lowest loss
6	Comprehensive GPU (VRU)	Four tasks, seq_len=15,000, 2x advantage at extreme scale
7	Why It Works (VRU)	Spectral probe — W_h converges to $2/\pi$, $\phi^2/2 = 8/\pi^2$
8	Probe Journey (VRU)	Five probes — spectral insulation finding, 7x rho stability
9	Arithmetic Benchmark (VRU)	L3 plateau diagnosis, scratchpad hypothesis
10+	Ongoing	v11 scratchpad → v23 explicit registers

Table 3. Full document series. Document 2 sits between geometric origin and experimental validation.

Core finding of this document: $\phi = 4/\pi$ is not a hyperparameter. It is a geometric constant derived from the proportional geometry of the human figure, confirmed as a field attractor in simulation, and validated as a gradient stabilizer at $\text{seq_len}=1000$ with 1.57x gradient advantage over vanilla at identical parameter count. The four conjectures stated here are the testable claims that Documents 3–9 address.

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$\phi = 4/\pi = 1.273239\dots$ | derived from Da Vinci's Vitruvian Man | gradient stabilizer at arbitrary sequence length